

CARTILAGINOUS FISH

These fish have a skeleton of pliable cartilage rather than the hard bone found in most other vertebrate groups. Most are predators with acute senses.

Sharks, rays, and chimaeras are all cartilaginous fish although chimaeras have distinct anatomical differences from the other two groups. In chimaeras, the upper jaw is fused to the braincase and so cannot be moved independently. They also have teeth that grow continuously. In contrast, sharks regularly lose their hard enamel-covered teeth but replace them from extra rows lying flat behind the active teeth – a characteristic that helps make sharks some of the most formidable predators on Earth. The skin of all three groups – sharks, rays, and chimaeras – is protected by tooth-like scales called dermal denticles.

SEARCHING FOR PREY

The ocean is home to most cartilaginous fish and while many live on the seabed, the larger predatory shark species and plankton feeders roam in open water. The bullshark and over 100 other species can enter estuaries and swim up rivers and a few

live entirely in rivers. Most open-water cartilaginous fish are continually on the move because, unlike bony fish, they do not have a gas-filled swim bladder to maintain neutral buoyancy and therefore may sink if they stop swimming. The whale shark and other surface-feeding species have a large oily liver to prevent this. Predatory sharks are well known for their amazing ability to smell blood and home in on wounded fish and mammals. Cartilaginous fish also have the ability to detect the weak electrical fields that surround living creatures and while not unique to them, this sense is developed to an extraordinary extent in this group.

REPRODUCTION

All cartilaginous fish mate and have internal fertilization. Chimaeras and many smaller sharks and rays lay eggs, each protected by a tough egg capsule and commonly known as a “mermaid’s purse”. However, around sixty per cent of cartilaginous fish give birth to well-developed live young that are nourished in the uterus by egg yolk or by a placental connection to the mother. Unlike mammals, the young are independent from birth and neither parent takes any interest in their offspring.

PHYLUM	CHORDATA
CLASSES	ELASMOBRANCHII HOLOCEPHALI
ORDERS	14–17
FAMILIES	57
SPECIES	1,338



A white shark, *Carcharodon carcharias*, shows the razor-sharp, replaceable rows of teeth that make it such a fearsome predator.

CHIMAERAS

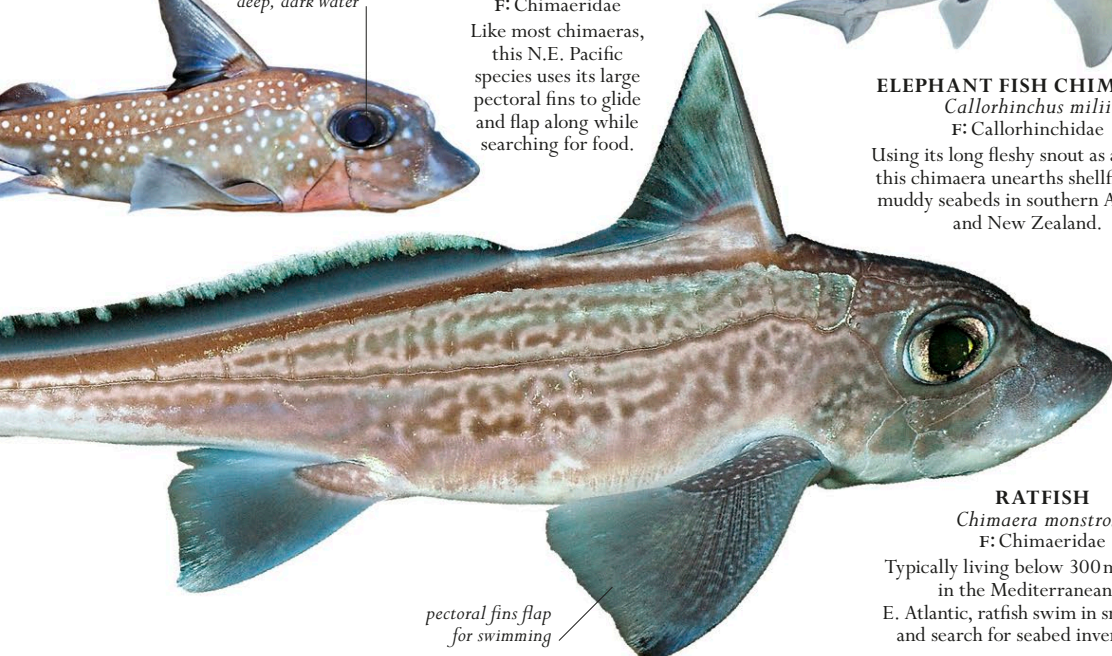
Also known as rabbitfish because of their fused plate-like teeth, the Holocephali are a small class of cartilaginous fish with one order, Chimaeriformes, and about 56 species. A strong, venomous spine in front of the first of two dorsal fins provides protection in their deepwater habitat.



large eye provides better vision in deep, dark water

SPOTTED RATFISH
Hydrolagus collieri
F: Chimaeridae

Like most chimaeras, this N.E. Pacific species uses its large pectoral fins to glide and flap along while searching for food.



pectoral fins flap for swimming

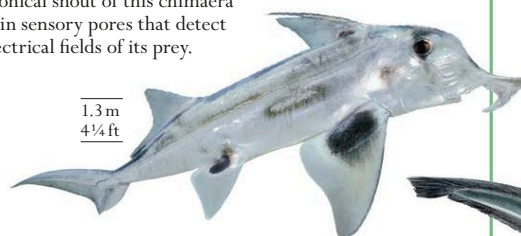
RATFISH
Chimaera monstrosa
F: Chimaeridae

Typically living below 300 m (1,000 ft) in the Mediterranean and E. Atlantic, ratfish swim in small groups and search for seabed invertebrates.



PACIFIC SPOOKFISH
Rhinochimaera pacifica
F: Rhinochimaeridae

The long, conical snout of this chimaera is covered in sensory pores that detect the electrical fields of its prey.



ELEPHANT FISH CHIMAERA
Callorhynchus milii
F: Callorhynchidae

Using its long fleshy snout as a plough, this chimaera unearths shellfish from muddy seabeds in southern Australia and New Zealand.

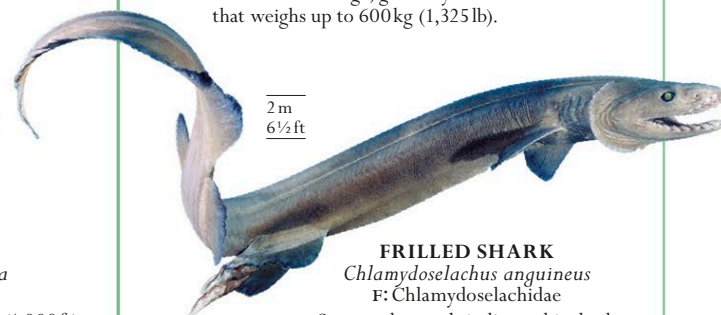
SIX- AND SEVEN-GILL SHARKS

Most sharks have five pairs of gill slits, but those in the order Hexanchiformes have either six or seven pairs. All six known species live in deep water. Two of them, called frilled sharks, have a long, soft, eel-like body and have recently been placed in their own order, Chlamydoselachiformes.



BLUNTNOSE SIXGILL SHARK
Hexanchus griseus
F: Hexanchidae

Rocky seamounts around the world are the haunt of this huge, green-eyed shark that weighs up to 600 kg (1,325 lb).



FRILLED SHARK
Chlamydoselachus anguineus
F: Chlamydoselachidae

Scattered records indicate this shark has a worldwide distribution. It has brilliant white teeth that may attract its prey of fish and squid.